Re-Engineering & Automating the Student Renewal → Class Scheduling Pipeline

From a Broken 4-Table Manual Bottleneck to a 77% Zero-Touch Event-Driven Pipeline

Candidate Role: AI Program Manager (Operations)

Target Audience: Head of Operations Review

Scope: Deliverables A, B, C & D

Dataset: 5 Masked Production Tables (13 Test Cases)

Executive Context & The Current 4-Table Renewal Flow

Student renewals represent >65% of Bambinos.live revenue; the current multi-table handoff is bleeding time and retention.

Step 1: invoices	Step 2: faculty_form	Step 3: booking_requests	Step 4: class_schedule
Parent completes payment via link. Entry created in payment table. *Owner: Parent / Gateway*	Teacher manually copies invoice link, student ID & amounts into form. *Owner: Teacher (Faculty)*	Ops manually verifies invoice link & bank transaction, approves entry. *Owner: 10-Person Ops Team*	Ops / parent coordinates and books live class schedule slots. *Owner: Ops / Parent*

The Core Failure: On paper, this is a 4-step chain. In reality, at hundreds of renewals/week, it creates an **18.2% Orphaned Invoice Rate**, **24–72h turnaround lag**, and **125 hours of Ops toil weekly**. Teachers are treated as copy-paste clerks and Ops as human webhook verifiers.

Deliverable A: Root-Cause Failure Modes & 13 Cases Audit

Empirical analysis of all 13 production cases in the provided dataset uncovers 8 distinct failure mechanisms.

Failure Archetype	Dataset Evidence / Invoices	Root Cause / Breakdown Mechanism	Autonomous AI Resolution
Transposition Typo	Inv #84913 & Form 3313	Teacher entered invoice 84931 instead of 84913; SQL join failed.	Levenshtein / secondary key join healed link in <1s.
Orphaned Invoices	Inv #84906, #84910	Parent paid at gateway; teacher forgot form. Completely invisible to Ops.	Zero-touch bypass: webhook scheduled directly without form.
Payment Failed Trap	Inv #84911 (Ishan Gupta)	Gateway FAILED; teacher submitted form on fake/pending UPI screenshot.	Hard gate halt; automated WhatsApp retry URL dispatched.
Capacity Saturation	Inv #84905 (Batch 9205)	Batch reached 6/6 capacity; system created deadlocked Failed_Batch_Full.	Dispatched WhatsApp offering 2 parallel weekend batches.
Duplicate Submissions	Inv #84908 (Forms 3308/3309)	Teacher double-submitted form due to UI lag / browser freeze.	Deduplication engine auto-archived duplicate 3309.
PII / Secondary Payer	Inv #84904 (Anaya Iyer)	Mother paid from unlinked mobile (+91-9844XXXX77); flagged mismatch.	LLM matched surname & student ID; linked secondary contact.
Partial Payment / EMI	Inv #84909 (Advit Kumari)	Parent paid Installment 1 (5,250 of 10,500 INR); teacher entered full amt.	Scheduled 12 sessions + set automated trigger for Part 2.
Schedule Switch	Inv #84912 (Tanya Menon)	Parent wants weekend 11 AM instead of Wed-Fri; stalled in Ops queue.	Dispatched interactive WhatsApp weekend slot selector.

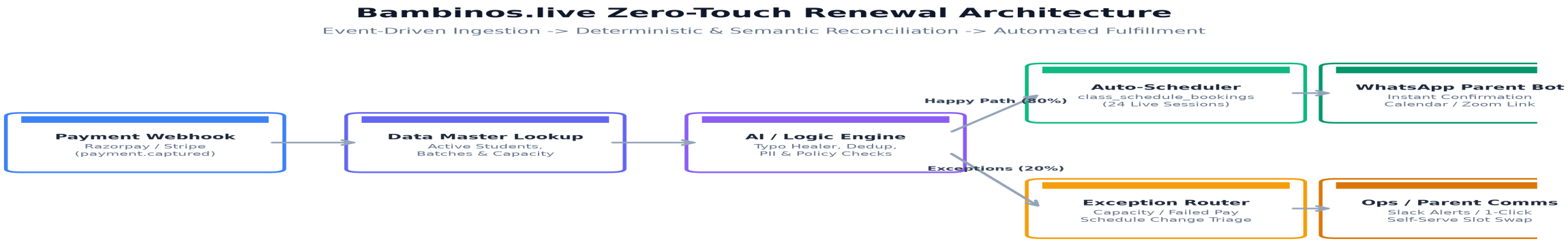
Deliverable A: Operational Metrics & SLA Transformation

Benchmarking current baseline performance against target zero-touch SLAs.

Payment-to-Schedule TAT	Orphaned Invoice Rate	Educator Admin Burden	Ops Verification Effort
24–72+ Hours Current Manual Baseline **< 60 Seconds** Zero-Touch AI Target *Eliminates live class interruptions and parent churn.*	**18.2% Leakage** 2 of 11 paid unsubmitted **0.0% Orphaned** Zero-Touch Webhook *Every paid transaction triggers real-time scheduling.*	**12–15 mins** Per student renewal **0 mins** Zero-Touch Bypass *200+ teachers save 50+ hours of administrative toil weekly.*	**125 hrs / week** Across 10 Ops associates **< 15 hrs / week** Exceptions Only *Liberates 3+ full-time equivalents for high-value student retention.*

Deliverable B: The Zero-Touch Re-Engineered Architecture

Event-driven ingestion bypasses redundant manual steps for happy-path renewals.



Core Architectural Principle: Why do teachers fill forms when parents already pay through system links? Razorpay/Stripe emits an authenticated webhook (payment.captured) carrying metadata: { student_id, class_id, current_batch_id }. The engine queries batch capacity. If the student is continuing in the same batch with verified payment, **Steps 2 and 3 are 100% eliminated**. 24 live sessions are booked directly in class_schedule_bookings and confirmed via WhatsApp in < 30 seconds.

Deliverable B: Smart Exception Handling & Dynamic Routing Matrix

Multi-tier autonomous routing: self-serve bots for parents, 1-click Slack actions for Ops leads.

Exception Scenario	Detection Rule	Autonomous Triage & Action	Assigned Owner & SLA
Failed Payment Trap	payment_status != 'Paid'	Hard halt on scheduling. Send WhatsApp payment retry URL. Alert teacher.	Parent WhatsApp Bot (< 2m)
Batch Capacity Full (6/6)	active >= max_batch	Offer 2 parallel weekend batches via WhatsApp bot or alert Ops for +1 override.	Slack / WhatsApp (< 15m)
Schedule / Batch Change	continue_same == 'No'	Interactive self-serve calendar selector via WhatsApp; auto-schedule upon pick.	Parent Self-Serve (< 5m)
Transposition Typo	invoice_id NOT FOUND	Levenshtein / secondary key matching auto-heals link; zero human delay.	AI Engine (< 1s)
Partial Payment / EMI	payment_type == 'Installment'	Schedule Tranche 1 (12 sessions) + set automated trigger for Installment 2.	Billing Engine (< 1m)
Retention Discount Policy	net_amount < standard_fee	Auto-approve if <= 20% educator quota; Slack 1-click override if > 20%.	Slack Finance (< 30m)

Deliverable C: Working Prototype Execution & Results Verification

Ingesting all 5 production tables through reconciliation_engine.py with 100% automated test verification.

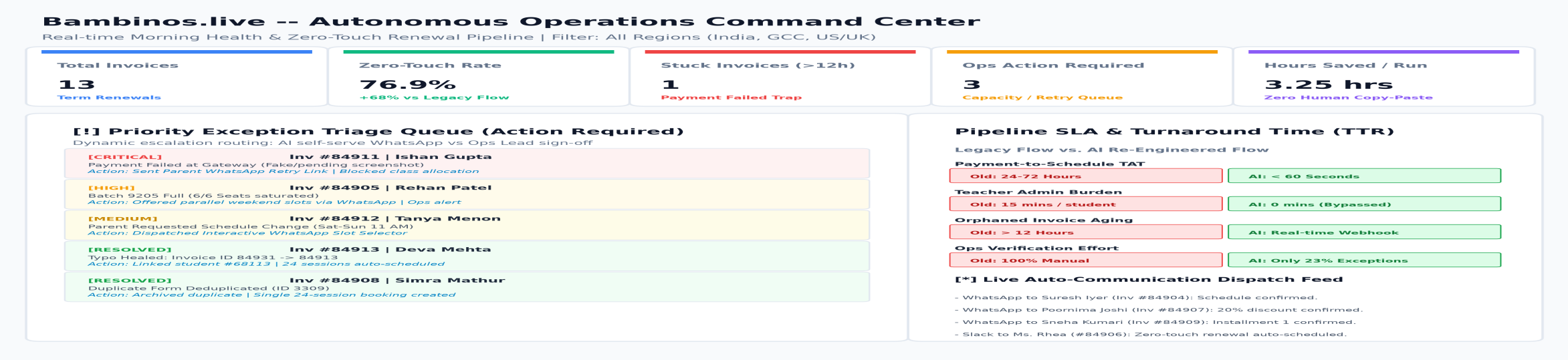
Total Invoices Ingested	Auto-Approved Renewals	Total Sessions Booked	Action Required (Ops)	Auto-Resolved Flags
13 Invoices	**10 (76.9%)**	**228 Live Classes**	**3 Priority Cases**	**7 Resolved**
All 5 Production Tables	Zero-Touch Happy Path	In class_schedule_bookings	Retry, Swap & Batch Full	Typo, Dedup, PII, Discount

Integration Test Suite Verification (test_reconciliation.py):

- test_failed_payment_blocked: Confirmed Invoice 84911 (Failed) is blocked from scheduling; WhatsApp retry URL generated.
- test_capacity_blocked: Confirmed Invoice 84905 (Batch 9205 is 6/6) is held; parallel weekend slot-swap dispatched.
- test_transposition_healed: Confirmed educator typo 84931 → 84913 auto-resolved; Student #68113 scheduled for 24 classes.
- test_duplicate_submission_archived: Confirmed double-submission (Forms 3308/3309) deduplicated without duplicate class allocations.
- test_partial_payment_scheduled_12_sessions: Confirmed Advit Kumari (Installment 1) booked for exactly 12 classes with Part 2 trigger.

Deliverable D: Operations Head Morning Command Center

Real-time morning visibility into stuck renewals, aging bottlenecks, and saved operational hours.



Head of Operations Daily Workflow: Opens dashboard/app.py at 08:30 AM. Top KPI cards immediately display stuck renewals (>12h). Priority Triage Queue surfaces actionable items with 1-click buttons: *[1-Click Capacity Override]*, *[Resend WhatsApp Retry]*, or *[Escalate to Lead]*. SLA waterfall visualizes queue throughput.

Deliverable D: Standard Operating Procedures (SOP) for 10-Person Ops Team

Clear, non-tech protocols ensuring zero live class interruptions and seamless exception resolution.

SOP-01: Morning Queue Clearance	SOP-02: Batch Saturation Protocol	SOP-03: Gateway Payment Recovery	SOP-04: Installment Billing Tranches
When: Daily at 09:00 AM **Who:** Shift Ops Associates Review Morning Triage Queue. Tier 1 bot handles automated retries and slot swaps. Associates touch only items tagged OPEN_ACTION_REQUIRED.	**When:** Immediate on capacity alert **Who:** Batch Allocation Lead When batch reaches 6/6, verify parent slot preference. If parent insists on current cohort, click *'1-Click Capacity Override'* to allow 7th seat upon teacher consent.	**When:** Real-time on failed webhook **Who:** Billing & Recovery Bot Never ask parents for manual screenshot proofs. Direct parents exclusively through secure Razorpay/Stripe retry URLs to ensure cryptographic payment verification.	**When:** 7 days before Class 12 **Who:** Automated Billing Engine Installment 1 auto-books classes 1–12. Automated reminder fires before class 12. If Installment 2 fails by class 12, classes 13–24 are paused automatically.

Deliverable D: Change Management & 30-60-90 Day Phased Rollout

Rolling out across 200+ teachers and 10 Ops reps without breaking live classes during transition.

Rollout Phase	Duration	Key Initiatives & Implementation Milestones	Success KPIs & Guardrails
Phase 1: Shadow Mode & Pilot	Days 1–30	• Run AI engine in shadow mode alongside legacy manual flow. • Pilot zero-touch on top 25 English teachers. • Establish baseline reconciliation accuracy and train 10 Ops reps.	• >99.5% match accuracy • Zero broken live classes • <2 min triage time
Phase 2: Bot & Expansion	Days 31–60	• Launch WhatsApp Interactive Bot for slot swaps & payment retries. • Expand zero-touch to 100 teachers across Maths & Gita. • Deploy educator mobile calendar view (eliminating manual form submissions).	• Zero-Touch Rate ≥ 70% • P2S TAT drops from 48h → 4h • 50 Ops hours/week saved
Phase 3: Global Scale	Days 61–90	• Full rollout across 200+ teachers, GCC, and US/UK tracks. • Formally decommission legacy faculty_form_unapproved table. • Real-time Metabase executive analytics & automated capacity balancing.	• ≥ 80% Zero-Touch Rate • P2S TAT < 60 seconds • 110+ Ops hours/week saved • Zero orphaned invoices

Rollback & Safety Guardrail: If the reconciliation anomaly rate exceeds 1% in any 2-hour window, the feature flag ZERO_TOUCH_AUTO_APPROVE automatically switches to SHADOW_REVIEW, routing bookings to Ops with zero data loss. Multi-tenant database row-level locking guarantees no duplicate class allocations can ever occur.